13 Percentiles: Cumulative Distribution Functions and Quantile Functions

- The probability density function of a continuous random variable describes which values are more or less likely in the "close to" sense.
- However, the pdf does not provide probabilities directly; rather, it is areas under the pdf which provide probabilities.
- There are many ways of describing distributions. In particular, a distribution is basically just a collection of percentiles.
- Cumulative distribution functions and quantile functions describe distributions by specifying all the percentiles.
- Roughly,
  - The *cumulative distribution function (cdf)* of a random variable fills in the blank for any given x: x is the (blank) percentile.
  - The *quantile function* fills in the following blank for a given $p \in (0, 1)$: the $(100p)$th percentile is (blank).

Example 13.1. Explain what the percentiles mean in the following statements.

- 47cm is 12th percentile of heights for newborn girls.
- 620 is the 80th percentile of SAT Math scores.

- A distribution is characterized by its percentiles.
- Roughly, the value x is the p-th **percentile** (a.k.a. quantile) of a distribution of a random variable X if p percent of values of the variable are less than or equal to x: $\mathrm{P}(X \le x) = p$.
- A few commonly used percentiles
  - First (or lower) quartile is the 25th percentile ($p = 0.25$)
  - Median is the 50th percentile ($p = 0.50$)

- Third (or upper) quartile is the 75th percentile ($p = 0.75$)

Example 13.2. Recall Example 12.1 and Example 12.2 where office arrival times, measured in minutes after 8am, followed this pdf

$$f_X(x) = \begin{cases} (2/3600)x, & 0 < x < 60, \\ 0 & \text{otherwise.} \end{cases}$$

1. Compute $\mathrm{P}(X < 15)$ and phrase the result in terms of a percentile.

2. Compute $\mathrm{P}(X < 45)$ and phrase the result in terms of a percentile.

3. Find an expression for $\mathrm{P}(X \leq x)$ for general x. This function, denoted $F_X(x)$, is called the *cumulative distribution function (cdf)* of X.

4. How does the cdf relate to percentiles?

5. Use the cdf to determine what percentile 30 minutes after 8am is.

6. Compute and interpret $F(40) - F(30)$.

- The *cumulative distribution function (cdf)* of a random variable fills in the blank for any given x: x is the (blank) percentile. That is, for an input x, the cdf outputs $\mathrm{P}(X \le x)$.
- The **cumulative distribution function (cdf)** of a random variable X is the function F_X defined by $F_X(x) = \mathrm{P}(X \le x)$. A cdf is defined for all real numbers x regardless of whether x is a possible value of X.

Example 13.3. Continuing Example 13.2 where X (minutes after 8am) has pdf

$$f_X(x) = (2/3600)x, \qquad 0 < x < 60.$$

and cdf

$$F_X(x) = \begin{cases} 0, & x < 0, \\ (x/60)^2, & 0 < x < 60, \\ 1, & x > 60. \end{cases}$$

1. Find and interpret the 10th percentile.

2. Find and interpret the 90th percentile.

3. Find an equation for the $(100p)$th percentile (e.g., $p = 0.9$ for the 90th percentile).

- The *quantile function* (essentially the inverse cdf) fills in the following blank for a given $p \in (0, 1)$: the $100p$th percentile is (blank).
- For example, evaluating the quantile function at $p = 0.25$ outputs the 25th percentile.
- For a *continuous* random variable X with cdf F_X, the **quantile function** Q_X is the inverse of the cdf, $Q_X(p) = F_X^{-1}(p)$.
- A quantile function basically gives you the same information as a cdf, just in the other direction.

- Both a quantile function and a cdf define a distribution by specifying all the percentiles.

Example 13.4. Continuing Example 13.3 where X (minutes after 8am) has pdf

$$f_X(x) = (2/3600)x, \qquad 0 < x < 60.$$

and cdf

$$F_X(x) = \begin{cases} 0, & x < 0, \\ (x/60)^2, & 0 < x < 60, \\ 1, & x > 60. \end{cases}$$

and quantile function

$$Q(p) = 60\sqrt{p}, \qquad 0 < p < 1.$$

1. Construct a circular spinner for simulating values of X. Version 1: label the values 0, 10, 20, 30, 40, 50, 60. Careful: they will not be equally spaced.

2. Sketch a histogram with equal bin widths—0 to 10, 10 to 20, etc.—that you would expect to see after spinning the version 1 spinner many times. Does it have the proper shape?

3. Construct a circular spinner for simulating values of X. Version 2: label the circular axis so that your spinner has 12 sectors each with probability 1/12.

4. Sketch a histogram that you would expect to see after spinning the version 2 spinner many times. Start by sketching a histogram with *unequal* bin widths, then sketch the correspond equal bin width histogram. Does it have the proper shape?

- The cdf and the quantile function can be used to create a spinner for a distribution.
- A spinner basically describes a distribution by specifying all the percentiles. For example,
  - the 25th percentile goes 25% of the way around ("3 o clock")
  - the 50th percentile goes 50% of the way around ("6 o clock")
  - the 75th percentile goes 75% of the way around ("9 o clock")
- Intervals corresponding to regions of higher density ("more likely" values) are stretched out on the spinner boundary; intervals corresponding regions of lower density ("less likely" values) are shrunk.
- **Universality of the Uniform (or "one spinner to rule them all").** Let F be a cdf and Q its corresponding quantile function. Let U have a Uniform(0, 1) distribution and define the random variable $X = Q(U)$. Then the cdf of X is F.
- Universality of the uniform might look complicated but all it basically says is that you can construct a spinner by putting the 25th percentile 25% of the way around, the 75th percentile 75% of the way around, etc.
- Actually, universality of the uniform says we don't have to create a new spinner. We can just spin the Uniform(0, 1) spinner, with evenly spaced values from 0 to 1, and transform each resulting value by plugging it into the quantile function.

Example 13.5. Continuing Example 12.3. Suppose that X is a claim amount in excess of the decuctible for a car insurance policy that incurs a claim, measured in thousands of dollars. Assume that X has pdf

$$f_X(x) = (1/4.3)e^{-x/4.3}, \qquad x > 0$$

1. Compute $\mathrm{P}(X < 5)$ and phrase the result in terms of a percentile.

2. Find an expression for the CDF of X.

3. Compute the 25th percentile.

4. Find an expression for the quantile function of X.

5. Use the quantile function to find the 75th percentile.

13.1 Exercises

Exercise 13.1. Continuing Exercise 12.1. In a certain population, household income X ($ thousands) follows the pdf

$$f_X(x) = cx^{-2.5}, \quad x \geq 30$$

for an appropriate constant c. Note that the measurement units are $ thousands, so 1 represents 1 thousand dollars. (This is not a super realistic example, but just go with it.)

1. Compute $\mathrm{P}(X \leq 100)$ and phrase the result in terms of a percentile.

2. Find the cdf of X.

3. Find the 10th percentile.

4. Find the quantile function of X.

5. Sketch a circular spinner for simulating values of X.

Exercise 13.2. Let X be the number of heads in 3 flips of a fair coin.

1. Find the pmf of X and sketch a plot of it.

2. Find the cdf of X and sketch a plot of it.

3. Let Y be the number of tails in 3 flips of a fair coin. Find the cdf of Y.

- A cdf is a non-decreasing function: if $x_1 \leq x_2$ then $F_X(x_1) \leq F_X(x_2)$.
- A cdf approaches 0 as the input approaches $-\infty$: $\lim_{x\to-\infty} F_X(x) = 0$
- A cdf approaches 1 as the input approaches ∞: $\lim_{x\to\infty} F_X(x) = 1$
- The cdf of a *discrete* random variable is a step function.
  - The steps occur at the possible values of the random variable.
  - The height of a particular step corresponds to the probability of that value, given by the pmf.
- The cdf of a *continuous* random variable is a continuous function.
  - The cdf of a *continuous* random variable is obtained by integrating the pdf, so

- The pdf of a *continuous* random variable is obtained by differentiating the cdf

$$F'_X = f_X \qquad \text{if } X \text{ is continuous}$$

- For any random variable X with cdf F_X

$$F_X(b) - F_X(a) = \mathrm{P}(a < X \leq b)$$

  Whether the inequalities in the above event are strict ($<$) or not ($\leq$) matters for discrete random variables, but not for continuous.
- Random variables X and Y **have the same distribution** if their cdfs are the same, that is, if $F_X(u) = F_Y(u)$ for all u.
- That is, two random variables have the same distribution if all the percentiles are the same.